Lecture on Model Predictive Control
Technical proofs

Laurent Pfeiffer

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Proof of Lemma 1. Let $x \in \mathbb{Y}_N$. By definition, there exists $\bar{u} \in \mathbb{U}^N(x)$. Set $y = y[\bar{u}, x]$. We have

$$\bar{u}(k) \in \mathbb{U}(x(k)), \quad y(k) \in \mathbb{Y}, \quad \forall k = 0, \dots, N-1, \quad y(N) \in \mathbb{Y}_0.$$

By assumption, there exists $u \in \mathbb{U}(y(N))$ such that $f(u, y(N)) \in \mathbb{Y}_0$. Next we define $\tilde{u} \in U^{N+1}$ as the concatenation of $\bar{u}$ and u, that is to say,

$$\tilde{u}(k) = \bar{u}(k), \quad \forall k = 0, \dots, N-1, \quad \tilde{u}(N) = u.$$

Clearly $\tilde{u} \in \mathbb{U}^{N+1}(x)$. This shows that $x \in \mathbb{Y}_{N+1}$ and finally that $\mathbb{Y}_N \subseteq \mathbb{Y}_{N+1}$. □

Proof of Theorem 5. Step 1: first properties. Let α and $\alpha' \in \mathcal{K}$. It is easy to verify that (i) $\alpha \circ \alpha' \in \mathcal{K}$ (ii) if α and $\alpha' \in \mathcal{K}_\infty$ then $\alpha \circ \alpha' \in \mathcal{K}_\infty$, and (iii) if $\alpha \in \mathcal{K}_\infty$ then α is a bijection from $[0, \infty)$ to $[0, \infty)$ and $\alpha^{-1} \in \mathcal{K}_\infty$.

Step 2: estimation of the decay. We have $\alpha_2^{-1} \in \mathcal{K}_\infty$. We deduce from the inequality $V(y) \leq \alpha_2(|y|)$ that

$$\alpha_2^{-1}(V(y)) \leq |y| \quad \text{and} \quad \alpha_3 \circ \alpha_2^{-1}(V(y)) \leq \alpha_3(|y|).$$

From the decay condition $V(g(y)) \leq V(y) - \alpha_3(|y|)$, we deduce that

$$V(g(y)) \leq h(V(y)), \tag{1}$$

where the function $h \colon [0, \infty) \to [0, \infty)$ is defined by

$$h(r) = r - \alpha_3 \circ \alpha_2^{-1}(r).$$

Note that $h \geq 0$ since by assumption $V \geq 0$. We also have $\alpha_3 \circ \alpha_2^{-1}(r) \geq 0$ and $\alpha_3 \circ \alpha_2^{-1}(r) = 0 \iff r = 0$. Therefore $h(r) \leq r$, for any $r \geq 0$, moreover, $h(r) = r$ if and only $r = 0$.

Step 3: regularization of h. We define the function $\tilde{h} \colon [0, \infty) \to [0, \infty)$ by

$$\tilde{h}(r) = \sup_{r' \in [0,r]} \Big\{ h(r') + \frac{r - r'}{2} \Big\}.$$

The following properties hold true:

1. For all $r \geq 0$, $\tilde{h}(r) \geq h(r)$, $\tilde{h}(r) \geq r/2$, and $\tilde{h}(r) \leq r$.

2. The function $\tilde{h}$ is strictly increasing.

3. For all $r \geq 0$, $h(r) = r$ if and only if $r = 0$.

4. The function $\tilde{h}$ is continuous.

The verification of the first two claims is left to the reader. Let us prove the third claim. We have $h(0) = 0$. Let $r \geq 0$ be such that $\tilde{h}(r) = r$. By continuity of h, there exists r' such that $\tilde{h}(r) = h(r') + (r - r')/2$. It follows that

$$h(r') - r' = \left(\tilde{h}(r) - \frac{r - r'}{2}\right) - r' = r - \frac{r - r'}{2} - r' = \frac{r - r'}{2} \geq 0.$$

But we also have $0 \geq h(r') - r'$, so necessarily $h(r') = r'$ and $r = r'$. As was seen in the second step of proof, $h(r') = r'$ implies that $r' = 0$, so finally $r = r' = 0$. The third claim is proved.

Let us prove the continuity of $\tilde{h}$. Let $r \geq 0$. Let $(r_n)_{n\in\mathbb{N}}$ be a convergent sequence with limit r. Let r' and r'_n be maximizers in the definitions of $\tilde{h}(r)$ and $\tilde{h}(r_n)$, respectively. Extracting a subsequence if necessary, $(r'_n)_{n\in\mathbb{N}}$ convergences to a limit point $s \leq r$. This implies that

$$\tilde{h}(r) \geq h(s) + \frac{r - s}{2} = \lim_{n\to\infty} h(r'_n) + \frac{r_n - r'_n}{2} = \lim_{n\to\infty} \tilde{h}(r_n).$$

Now, let us define $s_n = \min(r', r_n)$. Since $r' \leq r$ and $r_n \to r$, it holds that $s_n \to r'$. Therefore we have

$$\tilde{h}(r) = h(r') + \frac{r - r'}{2} = \lim_{n\to\infty} h(s_n) + \frac{r_n - s_n}{2} \leq \liminf_{n\to\infty} \tilde{h}(r_n).$$

The continuity of h follows.

Step 4: compositions of $\tilde{h}$. We define by induction the sequence of functions $(\tilde{h}^k)_{k\in\mathbb{N}}$, defined from $[0, \infty)$ to $[0, \infty)$ by $h^0 = h$ and $h^{k+1} = h \circ h^k$, for all $k \in \mathbb{N}$. We claim that for all $r \geq 0$,

$$\tilde{h}^k(r) \underset{k\to\infty}{\longrightarrow} 0. \tag{2}$$

The sequence $(\tilde{h}^k(r))_{k\in\mathbb{N}}$ is decreasing and bounded from below, therefore it converges. Since $\tilde{h}$ is continuous, the limit value is a fixed point of $\tilde{h}$. But the only fixed point of $\tilde{h}$ is 0, which proves (2).

Step 5: conclusion. From (1), we deduce that $V(g(y)) \leq \tilde{h}(V(y))$. It follows, by induction, that

$$V(y[x](k)) \leq \tilde{h}^k(V(y)).$$

We next infer that

$$|y[x](x)| \leq \beta(|x|, k),$$

where $\beta(r, k) = \alpha_1^{-1} \circ \tilde{h}^k \circ \alpha_2(r)$. For a given k, we have that $\beta(\cdot, k) \in \mathcal{K}$ as a composition of $k + 2$ functions in $\mathcal{K}$. We also have $\beta(r, k) \to 0$ as $k \to \infty$, as a consequence of (2), the continuity of the functions α_1^{-1} and α_2, and the fact that $\alpha_2(0) = \alpha_1^{-1}(0) = 0$. The theorem is proved. □

Lemma 1 (Abstract dynamic programming principle)**.** *Let $\mathcal{X}$ and $\mathcal{Y}$ be two sets. Let $f_1 \colon \mathcal{X} \to \mathbb{R}$ and $f_2 \colon \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$. Consider the problem*

$$\inf_{(x,y)\in\mathcal{X}\times\mathcal{Y}} \; f_1(x) + f_2(x,y). \tag{P}$$

Consider the problem

$$\inf_{x\in\mathcal{X}} \; f_1(x) + V(x) \tag{P_1}$$

where the function $V \colon \mathcal{X} \to \mathbb{R} \cup \{-\infty\}$ is defined by

$$\inf_{y\in\mathcal{Y}} \; f_2(x,y). \tag{$P_2(x)$}$$

Then Problems (P) and (P_1) have the same optimal cost. Moreover, for all $(x,y) \in \mathcal{X}\times\mathcal{Y}$, (x,y) is a solution to Problem (P) if and only if x is a solution to Problem (P_1) and y is a solution to Problem ($P_2(x)$).

Proof. We denote as $\mathrm{val}(P)$ and $\mathrm{val}(P_1)$ the optimal costs of Problems (P) and (P_1). It holds that

$$\begin{aligned}
\mathrm{val}(P) &= \inf_{(x,y)\in\mathcal{X}\times\mathcal{Y}} \Big\{ f_1(x) + f_2(x,y) \Big\} \\
&= \inf_{x\in\mathcal{X}} \Big\{ \inf_{y\in\mathcal{Y}} \Big\{ f_1(x) + f_2(x,y) \Big\} \Big\} \\
&= \inf_{x\in\mathcal{X}} \Big\{ f_1(x) + \inf_{y\in\mathcal{Y}} f_2(x,y) \Big\} \\
&= \inf_{x\in\mathcal{X}} \Big\{ f_1(x) + V(x) \Big\} \\
&= \mathrm{val}(P_1).
\end{aligned}$$

So Problems (P) and (P_1) have the same optimal cost. Let us denote this common value as v. We have

$$f_1(x) + f_2(x,y) - v = \big(f_1(x) + V(x) - v\big) + \big(f_2(x,y) - V(x)\big).$$

Note that $f_1(x)+V(x)-v \geq 0$ and $f_2(x,y)-V(x) \geq 0$. We obtain the desired equivalence:

$$\begin{aligned}
(x,y) \text{ is a solution to } (P) &\iff 0 = f_1(x) + f_2(x,y) - v \\
&\iff 0 = f_1(x) + V(x) - v \text{ and } 0 = f_2(x,y) - V(x) \\
&\iff x \text{ is a solution to } (P_1) \text{ and } y \text{ is a solution to } (P_2(x)).
\end{aligned}$$

This concludes the proof. □

Proof of Theorem 8. Let $x \in \mathbb{X}$, let u denote the control generated by the MPC method (for the time horizon N), and let y denote the associated state. For all $k \in \mathbb{N}$, we have $y(k+1) = f_N(y(k))$. Thus the (ADP) assumption writes

$$V_N(y(k+1)) \leq V_N(y(k)) - \alpha\ell(u(k), y(k)).$$

Rearranging, we obtain that

$$\ell(u(k), y(k)) \leq \frac{1}{\alpha}\big(V_N(y(k)) - V_N(y(k+1))\big).$$

Let $K \in \mathbb{N}$. We have

$$\begin{aligned}\sum_{k=0}^{K-1} \ell(u(k), y(k)) &\leq \frac{1}{\alpha} \sum_{k=0}^{K-1} \big(V_N(y(k)) - V_N(y(k+1))\big) \\ &= \frac{1}{\alpha}\big(V(x) - V(y(K+1))\big) \leq \frac{1}{\alpha} V(x).\end{aligned}$$

Passing to the limit as $K \to \infty$, we infer that

$$J_\infty(u, x) = \lim_{K \to \infty} \sum_{k=0}^{K-1} \ell(u(k), y(k)) \leq \frac{1}{\alpha} V(x).$$

The second part of the theorem is a rather direct application of Theorem 5. Let us set

$$\ell^*(x) = \inf_{u \in \mathbb{U}(x)} \ell(u, x).$$

By assumption, we have $\ell^*(x) \geq \alpha_3(|x|)$. It holds that

$$\alpha_3(|x|) \leq \ell^*(x) \leq V_N(x) \leq \alpha_2(|x|)$$

and

$$V_N(f_N(x)) \leq V_N(x) - \alpha \ell(\mu_N(x), x) \leq V_N(x) - \alpha \ell^*(x) \leq V_N(x) - \alpha \alpha_3(|x|).$$

Therefore the assumptions of Theorem 5 are satisfied and y^* is asymptotically stable. □

Proof of Lemma 9. Let $N > 0$ and let u denote a solution of $\mathcal{P}_N(x)$. Let y denote the associated trajectory. We have $y(N) \in \mathbb{Y}_0$. We extend the control sequence of one element defined by $u(N) = \kappa(y(N))$. We extend y correspondingly with $y(N+1) = f(u(N), y(N))$. We have $u(N) \in \mathbb{U}(y(N))$ and $y(N+1) \in \mathbb{Y}_0$, so the extended control sequence belongs to $\mathbb{U}_{N+1}(x)$. In addition, we have

$$\begin{aligned}J_N(u, x) &= \sum_{k=0}^{N-1} \ell(u(k), y(k)) + \underbrace{\ell(u(N), y(N)) + F(y(N+1))}_{\leq F(y(N))} \\ &\leq V_{N-1}(x).\end{aligned}$$

It follows that $V_{N-1}(x) \geq V_N(x)$. □

Proof of Corollary 10. The combination of Theorem 6 and Lemma 9 directly yields the (ADP) assumption with $\alpha = 1$. The result follows with Theorem 8. □

Proof of Lemma 11 (first part). Let $N \in \mathbb{N}$ and let $x \in \mathbb{Y}_{N+1}$. Let u be a solution to $\mathbb{P}_{N+1}(x)$. Denote as $\tilde{u}$ the restriction of u to its N first elements. Then it holds that

$$V_N(x) \leq J_N(\tilde{u}, x) \leq J_{N+1}(u, x) = V_{N+1}(x),$$

as was to be proved. The inequality $V_N(x) \leq V_\infty(x)$ can be proved similarly. □

Proof of Lemma 11 (second part). Step 1: dynamic programming principle. We state here (without proof) a quite direct extension of the dynamic programming principle seen in Theorem 6: For any $N \in \mathbb{N}$, for any $M \in \mathbb{N}$ such that $M \leq N$, for any $x' \in \mathbb{X}$, it holds that

$$V_N(x') = \inf_{u \in \mathbb{U}_M(x')} \left\{ \sum_{k=0}^{M-1} \ell(u'(k), y[u', x'](k)) + V_{N-M}(y[u', x'](M)) \right\}. \tag{3}$$

Theorem 6 corresponds to the special case where $M = 1$.

Step 2: a first estimate for the decay of V. We turn to the proof of the theorem. Let $x \in \mathbb{Y}_N$ and let u denote a solution to $\mathcal{P}_N(x)$. Let y denote the associated trajectory. We first prove the following inequality: for any $0 \leq M \leq N-1$,

$$V_N(y(1)) \leq V_N(x) - \ell(x, u(0)) + V_{N-M}(y(M+1)). \tag{4}$$

Note that this inequality is pretty similar to the desired (ADP) inequality. We first utilize (3), taking $x' = y(1)$ and $u' = (u(1), \ldots, u(M))$. We have $y[u', x'] = (y(1), \ldots, y(M+1))$. It follows that

$$V_N(y(1)) \leq \sum_{k=1}^{M} \ell(u(k), y(k)) + V_{N-M}(y(M+1)). \tag{5}$$

We also have the following inequality:

$$\sum_{k=1}^{M} \ell(u(k), y(k)) = \left(\sum_{k=0}^{M} \ell(u(k), y(k)) \right) - \ell(u(0), x) \leq V_N(x) - \ell(u(0), x). \tag{6}$$

Combining (5) and (6), we obtain the desired inequality (4). Note that at this stage, we did not make use of any structural assumption on the data.

Step 3: conclusion. At this stage, our main task is to find M such that $V_{N-M}(y(M+1))$ is small. Recall that

$$V_N(x) = \sum_{k=0}^{N-1} \ell(u(k), y(k)).$$

Let $k \in \{1, \ldots, N-1\}$ be such that

$$\ell(u(k), y(k)) \leq \frac{V_N(x)}{N-1}.$$

Let $M = k - 1$. We have

$$\begin{aligned} V_{N-M}(y(M+1)) &\leq V_N(y(k)) \leq \gamma \ell^*(y(k)) \leq \gamma \ell^*(u(k), y(k)) \\ &\leq \frac{\gamma}{N-1} V(x) \leq \frac{\gamma^2}{N-1} \ell^*(x) \leq \frac{\gamma^2}{N-1} \ell(u(0), x). \end{aligned}$$

Combining the above inequality with (4), we deduce that

$$V_N(y(1)) \leq V(x) - \left(1 - \frac{\gamma^2}{N-1}\right) \ell(u(0), x),$$

which concludes the proof. □